

Poverty Dynamics in Iran: Substitutes for Missing Data*

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Abstract

Objective: Measuring poverty dynamics—distinguishing chronic from transient poverty—is crucial for effective anti-poverty policies, yet constrained by limited panel data in developing countries.

Method: We overcome this constraint by constructing a synthetic panel from the Iranian Household Expenditure and Income Survey (HIES). Using a consumption model, we estimate first-round consumption for households only observed in the second round, enabling dynamic poverty analysis despite the absence of true longitudinal data.

Findings: Our analysis for 1398-1399 reveals that 29% of households were chronically poor (poor in both years). Chronic poverty is markedly higher among vulnerable subgroups: 54% in rural households, 40% in female-headed households, and 34% in households with non-university educated heads. Notably, our synthetic panel estimates consistently encompass the true panel values where available, validating the method's reliability.

Conclusion: Synthetic panels provide a viable alternative for tracking poverty transitions where traditional panel data are absent, revealing significant heterogeneities that warrant targeted policy attention.

Keywords: chronic and transient poverty, poverty dynamics, synthetic panel data

JEL Classification: XX; YY; ZZ

1 Introduction

Eradicating poverty is a crucial objective for nations worldwide, with the United Nations committing to "ending poverty in all its forms everywhere" as one of its

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sustainable goals. However, the COVID-19 pandemic has impeded progress towards this goal, resulting in an increase in the global rate of extreme poverty for the first time in over two decades and an additional 120 million people living in poverty in 2020¹.

The World Bank's standardized international poverty lineset at \$8.3 per person per day in 2021 Purchasing Power Parity (PPP) exchange rates has been customized for several nations, including Iran (World Bank, 2026). It is designed specifically to measure poverty at higher income levels and is particularly relevant for the economic context of upper-middle-income countries (Ferreira et al., 2016). This measure is a part of a larger tiered analytical framework that includes the revised extreme poverty line (\$3.00) and a lower-middle-income line (\$4.15), providing a multi-lens view of global deprivation, (Jolliffe et al., 2022).

In addition to the World Bank's poverty thresholds, a well-documented alternative approach to calculating the poverty headcount ratio uses the food expenditure share exceeding 50% of total consumption. This method is grounded in Engel's Law, which posits that as household income increases, the share of expenditure devoted to food declines (Engel, 1857). A share above 50% signals that a household is allocating the majority of its resources to basic subsistence, leaving little for other essential needs such as healthcare, education, and shelter (Ravallion, 1992, 2011). This indicator is behaviorally grounded, relatively inflation-resistant, and useful for cross-sectional and historical comparisons, especially in low-income settings (Deaton, 1997).

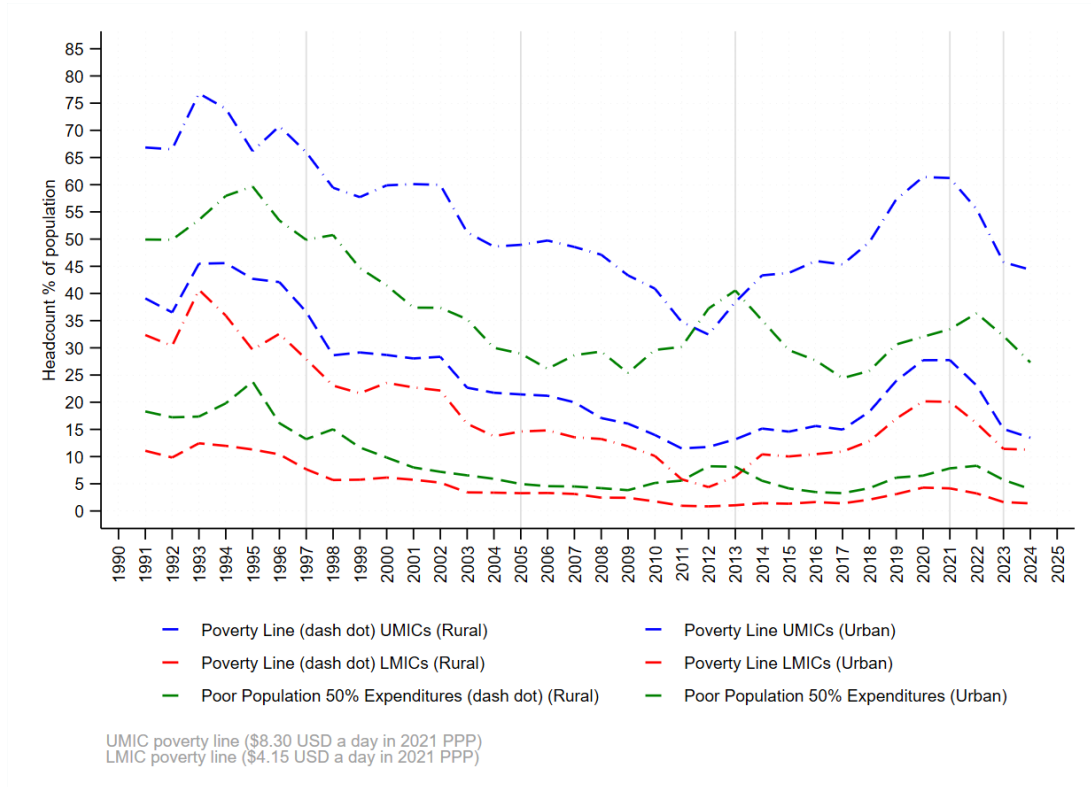
However, its application has significant limitations. It does not account for demographic differences, shifts in food quality, or the growing importance of non-food necessities in modern economies (Lipton, 1988; Banerjee and Duflo, 2011). Consequently, while informative, the 50% threshold is best employed as one component within a broader multidimensional poverty assessment, rather than as a standalone measure (Alkire and Foster, 2011; World Bank, 2017). Figure 1.1 plots the headcount ratio trends by region, level of poverty line upper-Middle-Income Countries (8.3 PPP), Lower-Middle-Income Countries (4.15 PPP) and the headcount ratio based on food expenditures over 50% of total monthly expenditures. The patterns of the trends are very similar and almost decreasing, although after the US exit from the JCPOA, the headcount ratio turned to exhibit increasing trend. The poverty rate in Rural areas is significantly higher than urban areas and in 2010 it reached to the highest level of about 50% in terms of 8.0 PPP exchange rate.

The utilization of the \$8.0 per day poverty line the global benchmark for Upper-Middle-Income Countries (UMICs) often yields significantly elevated poverty headcount ratios in rural contexts, suggesting a critical disconnect between the uniform international standard and local cost-of-living heterogeneity. This discrepancy is largely attributed to the non-application of intra-country spatial price deflators, which fail to account for systematic differences in the cost of a basic needs basket between urban and rural areas (Chen et al., 2020; Dabalen et al., 2019). Furthermore, the overestimation of rural poverty is compounded by the challenge of accurately valuing non-monetary consumptions specifically, own-produced food consumed by subsistence and semi-subsistence farming households which is a major component of real welfare in rural UMICs yet is frequently undervalued or inconsistently measured in household surveys (Ravallion, 2008).

Static poverty profiles and or national trends provide only a snapshot, conflating

¹<https://sdgs.un.org/goals/goal1>

Figure 1.1: Headcount Ratio, by Area of Residence



Note: LMICs and UMICs stand for lower-middle-income and upper-middle-income countries, respectively. All of the poverty lines reveal a quite similar trend for the population under the poverty line; however, the level of the lower-middle-income poverty line is arguably lower than the Engel coefficient and that for upper-middle-income countries. We use the consumption PPP converter produced by the World Bank, which is based on 2021 fixed prices. The gray vertical lines refer to presidential election years. The highest poverty rate was realized during Hashemi Rafsanjani's presidential administration, in the year that CPI growth exceeded 49 percent and triggered the first social unrest in some cities since the 1998 Islamic Revolution.

Source: HIES 1991-2024 for food, non-food and households' size, and world bank for PPP converter.

chronic and transitory poverty while obscuring welfare trajectories, duration, and vulnerability - limitations that undermine effective targeting [Baulch and Hoddinott \(2000b\)](#). Dynamic presentations, by contrast, disaggregate the poor into distinct groups, identify drivers of poverty transitions, assess policy effectiveness, and reveal risk and persistence [Barrett et al. \(2016\)](#). This richer understanding enables differentiated, evidence-based interventions that address both immediate needs and long-term structural vulnerabilities, making dynamic analysis indispensable for effective poverty reduction strategies [Carter and Barrett \(2006\)](#); [Dercon \(2005a\)](#); [Hulme and Shepherd \(2003\)](#). The shift from examining static poverty profiles or national trends to analyzing poverty dynamics constitutes a crucial methodological and theoretical advance in development economics and social policy. For our study of Iran, this transition to a dynamic perspective provides a richer understanding necessary for effective, targeted intervention. The core argument rests on the fact that poverty is an inherently heterogeneous and time-varying living circumstance ([Baulch and Hoddinott, 2000b](#)). Static analyses, typically derived from cross-sectional data, capture only a snapshot in time and are inherently limited in addressing the processes that determine an individual's or household's welfare trajectory. By contrast, a dynamic approach relying on longitudinal (panel) data allows for the disaggregation of the poor population into distinct sub-groups that static measures conflate ([Barrett et al., 2016](#)). The most significant conceptual contribu-

tion of poverty dynamics is the distinction between chronic and transitory poverty (Hulme and Shepherd, 2003).

Households or individuals who remain below the poverty line for an extended period often across multiple survey waves are classified as chronically poor. This group is typically trapped by long lasting structural challenges and limited asset endowments. In the Iranian context, for example, persistently high inflation, structural unemployment, and a low minimum wage (760 international PPP dollars per month, based on the implied PPP conversion rate of 289,230 for 2026) may be driving an increase in chronic poverty, as recent analyses suggest (Eslami and Hoseini, 2024). Households with transitory poverty move into and out of poverty, often due to temporary income shocks, such as illness, job loss, climate-related events (like floods), or volatile price fluctuations.

Policy prescriptions differ fundamentally for these two groups. Poverty alleviation strategies that rely on static data are inherently limited, as they apply generic interventions to a heterogeneous population, overlooking the distinct needs of chronic versus transitory poor.

A dynamic perspective enables scholars and policymakers to quantify and analyze vulnerability to poverty (Chaudhuri et al., 2002). This forward-looking metric identifies non-poor households that face a high risk of falling into poverty due to future shocks. Analyzing vulnerability is essential for designing preventative rather than purely reactive policy. In an environment like Iran characterized by macroeconomic volatility, high inflation, and devastating sanctions (World Bank, 2023)-understanding which households are vulnerable, particularly those in the shrinking middle class, becomes critically important for preventing a broader social catastrophe. Static analysis can only identify correlates of poverty, such as household size or the education level of the head. By contrast, dynamic analysis reveals the specific events and processes the drivers that cause households to transition into or out of poverty (McKay and Okidi, 2003). To avoid inclusion and exclusion error and to ensure efficient use of limited resources policymakers must distinguish between transient and chronic poverty, a task that requires dynamic analysis (Dang, 2020; Baulch and Hoddinott, 2000a).

Measuring movements in and out of poverty, particularly in developing countries, is a challenging task due to the lack of panel data, which is the most appropriate source of information. However, there is a growing body of literature that has developed techniques to measure poverty dynamics in the absence of panel data (Cruces et al., 2015).

This paper seeks to estimate poverty transitions in Iran using the "synthetic panel" method developed by (Dang et al., 2014), based on Expenditures and Income surveys taken in 2019 and 2020. These years are particularly important due to the severity of sanctions against Iran and the outbreak of the coronavirus. The synthetic panel method produces lower and upper bound estimates of poverty transitions, which are expected to sandwich true poverty transition estimates.

This paper is the first that validates the synthetic panel approach for a developing country under sanctions, and the results indicate that the lower and upper bounds fully encompass the true poverty dynamic measures in most cases. Additionally, the results show that almost 29% of urban and rural households were poor in both years, with higher rates in female-headed households, households with non-university educated heads, and rural households, indicating a need for special attention in these groups.

The rest of the paper is organized as follows. Section two outlines the under-

lying literature, establishing a theoretical and empirical foundation by reviewing the key distinction between static and dynamic poverty analysis and tracing the methodological evolution from panel data studies to synthetic panel techniques designed for contexts with repeated cross-sectional data. Section three details the data sources and descriptive statistics, while section four explicates the econometric methodology, focusing on the Dang, Lanjouw, Lanjouw, and Muñoz (DLLM) model for estimating poverty transition bounds. Section five presents the core empirical results, validating the synthetic panel estimates and analyzing poverty dynamics for the aggregate population and key socioeconomic subgroups. Section six discusses potential mechanisms driving the observed dynamics, and section seven concludes by summarizing the findings and their implications for policy and future research.

2 Literature

Poverty literature can be broadly categorized into two main approaches: poverty dynamics and static poverty analysis. While static poverty analysis is based on cross-sectional data from households, poverty dynamics studies utilize longitudinal data that tracks the same households over a period of time (Yaquub, 1999). This paper focuses specifically on the literature pertaining to poverty dynamics. In their work on poverty analysis, Dang (2020) has classified imputation methods based on the availability of consumption data into three categories: 1) completely missing, 2) partially missing, and 3) cross-sections available but missing panel data. In this paper, we will first review studies that have employed panel data for analyzing poverty dynamics, followed by an examination of alternative methods in line with Dang's categorization.

One notable example of a panel data study on poverty dynamics is the "Panel Study of Income Dynamics" project, initiated by Duncan et al. (1984) in 1968. The study surveyed 5,000 American households over a period of nearly ten years, covering various aspects of their income status and occupation, even if they had moved residences. The authors of the study also discussed different perspectives on poverty, such as poverty being caused by lack of skills and opportunities, or personal characteristics. They argue that poverty dynamics analysis can significantly improve our understanding of the extent and causes of poverty. Their findings indicate that between 1969 and 1978, around 25% of the population experienced transient poverty, while less than 1% were chronically poor.

Salehi-Isfahani and Majbouri (2013) conducted a study to estimate poverty dynamics in Iran between 1992 and 1995 using the Expenditures and Income Survey. They argue that using per capita household expenditures as a measure of welfare is more accurate than income, as it is derived from more questions and people tend to give inaccurate responses when asked about their income. They applied weights and averaged the expenditures over two consecutive years to correct for measurement errors and attrition. The study found that younger individuals and female-headed households are more likely to experience long-term poverty, and that an individual's place of residence can influence the duration of poverty. They conclude that providing education opportunities can be an effective strategy to reduce poverty, as it has a strong correlation with chronic poverty.

In their study, Lawson and McKay (2002) highlight the limitations of panel data in analyzing poverty dynamics, such as measurement errors that arise from track-

ing households over time and potential biases arising from interviewee reluctance. They argue that in the absence of panel data, alternative methods such as repeated cross-sections or proxies for chronic poverty can be used to estimate poverty dynamics. They also suggest that chronic poverty is typically caused by lack of human and physical assets, as well as location in remote or disadvantaged areas.

The problems associated with panel data, compounded by the lack of such data in developing countries, necessitate the exploration of alternative solutions for analyzing poverty dynamics. We continue to examine the literature on poverty dynamics according to the categories previously mentioned.

In the absence of panel data, researchers have proposed a variety of methods to estimate poverty dynamics. [Sahn and Stifel \(2000\)](#) for instance, propose a method for building a wealth index when consumption data are completely missing. They use DHS² data and various household characteristics, such as water source, toilet facilities, and construction materials, as well as durables, such as the ownership of a radio or television, to create a wealth index for 11 African countries between 1986 and 1998. Their findings suggest that the overall poverty rate has declined in this period, largely due to the growth of rural areas.

Another alternative method for estimating poverty dynamics in the absence of panel data is the use of a model to identify the factors affecting consumption, as proposed by [Mathiassen \(2013\)](#). This approach allows for the development of a light survey that can be collected annually, and when combined with consumption data, can be used to estimate the poverty rate.

The third category of literature, which includes the present research, consists of papers that use cross-sections to build panel data. [Dang et al. \(2014\)](#) propose a method, known as the DLLM method, for estimating poverty dynamics using two or more rounds of cross-sections in the absence of panel data. This method estimates a range for poverty dynamics by first creating a model of consumption using only time-invariant covariates in the first round of data, and then applying the same model to the second round of data. The authors provide both parametric and nonparametric bounds for Vietnam and Indonesia and compare the results with true values. They conclude that the bounds virtually always include the true panel estimates, and that the success of the method is contingent on the ability to predict the dependent variable of interest or capture the range of autocorrelation for the error terms.

[Cruces et al. \(2015\)](#) further validate the DLLM method in three countries and three different scenarios: when income is used as the measure of welfare instead of consumption, when the time interval between cross-sectional data is long, and when there are multiple transition lines. They choose Chile, Peru, and Nicaragua for their research due to the availability of panel data in these countries. They conclude that while the estimated bounds encompass the true value in different scenarios, the narrowness of the bounds is highly dependent on the selected variables.

It is worth noting that the DLLM method has its own limitations. One such limitation is the need for true panel data to estimate the correlation of error terms. To address this issue, [Dang and Lanjouw \(2013\)](#) propose a method that utilizes the country's data to estimate the correlation of error terms and obtain a point estimation instead of bounds. However, as highlighted by [Hérault and Jenkins \(2019\)](#), this method may not be suitable for developed countries, as the results differ significantly from the true values.

²Demographic and Health Surveys

As noted by [Bourguignon et al. \(2020\)](#), synthetic panels are highly dependent on the assumptions underlying the parameter estimation process. They propose an alternative method to estimate the correlation coefficient based on an AR(1) specification, and evaluate the results using real data from Mexico collected between 2002 and 2005. Although the results do not differ significantly from the real data, certain income mobility measures are highly sensitive to the value of the auto-regressive coefficient.

Poverty dynamics have also been studied in Iran. [Gerivani et al. \(2019\)](#) use data from the Expenditures and Income Survey of 2012, 2015, and 2016 to analyze poverty dynamics in urban areas. They find that the difference between the synthetic panel and the real panel is only 1.6 percent. Additionally, they find that 80 percent of the poor (non-poor) families in 1395 were also poor (non-poor) in 1391 and 1394.

Overall, these studies provide additional insights into the limitations of the DLLM method and alternative methods for estimating poverty dynamics in the absence of panel data. They also highlight the importance of considering the specific context and assumptions of the data and methods used in poverty dynamics analysis.

3 Data

In this paper, we utilize data from the Household Expenditures and Income Survey (HIES) for the years 2019 and 2020 in Iran. These years are particularly significant as they reflect the economic and social conditions in Iran during a time of increased poverty and macroeconomic instability, as a result of both the COVID-19 pandemic and US sanctions. According to the Poverty Monitoring report published by Iran's Ministry of Cooperatives, Labour, and Social Welfare in 2020, the poverty line rose by 38% from 2019 to 2020, reaching a level of 1,254,000 Toman.

The HIES is a nationally representative, two-stage stratified survey that has been conducted annually by the Statistical Center of Iran since 1935, with the exception of the years 1976, 1978 and 1981. The survey includes both demographic and income information, with a focus on expenditures. Information on over 1,000 items is collected from around 20,000 urban and 18,500 rural households in 2019, and 19,300 urban and 18,250 rural households in 2020. The survey consists of four sections: social characteristics of family members, place of residence and major household appliances, expenditures in the last month or 12 months for durable goods, and income of family members.

In this research, we extract households characteristics from section 1 and their expenditures from section 3. The poverty line used in this analysis is based on the Poverty Monitoring report and is set at 1,254,000 Toman per capita per month for 2020 and 906,000 Toman for 2019.

4 Model

We use the DLLM method to estimate poverty dynamics, which based on a consumption model is constructed for each year of 1 and 2:

$$C_{i1} = \beta'_1 X_{i1} + \epsilon_{i1} \quad (1)$$

$$C_{j2} = \beta'_2 X_{j1} + \epsilon_{j2} \quad (2)$$

Where, C is the variable of consumption, i and j represents the households, X_i is the vector of time-invariant covariates, β is the vector of parameters to be estimated, and ϵ is the error term. The consumption of year 1 for household j who is surveyed in year 2 is not observable, but it can be estimated. The two equations are estimated using Ordinary Least Squares (OLS) regression. The variables selected for the models are: Households head age, sex, number of years of education, number of children, area of living (urban/rural), province, households size, status of employment (employed/unemployed) and status of marriage (married/single) of the household's head.

The DLLM method makes two key assumptions: first, that the underlying population sampled is the same in both years, and second, that the error terms are positive quadrant dependent, meaning that the correlation of the error terms is non-negative. Based on these assumptions, we can derive nonparametric bounds. The upper bound estimates are given by the probability that the two error terms are completely independent of each other. To derive the nonparametric upper bounds, for each household j in round 2, a random draw with replacement is taken from the empirical distribution of the predicted residuals and the consumption of year 1 is estimated using the following equation:

$$\widehat{C}_{j1}^U = \hat{\beta}'_1 X_{j2} + \epsilon_{j1}$$

The lower bound estimates are given by the probability that the two error terms in the consumption model are completely dependent on each other. the consumption of year 1 is estimated using the following equation:

$$\widehat{C}_{j1}^L = \hat{\beta}'_1 X_{j2} + \gamma \hat{\epsilon}_{j2}$$

Where scalar γ is chosen to ensure the standard deviation of the imputed Year 1 residuals distribution equals $\hat{\sigma}_1$.

In order to narrow the bounds, DLLM proposed a parametric approach which assumes that the error terms have a bivariate normal distribution. The correlation between the error terms, denoted as ρ , is calculated using true panel data from other similar countries. This allows for the calculation of poverty dynamics statistics such as the probability of being poor in year 1 and also poor in year 2, denoted as $Prob(C_{ji} < z_1 \text{ and } C_{j2} < z_2)$, using the bivariate normal cumulative distribution function, denoted as ϕ , and the poverty lines for year 1 and year 2, denoted as z_1 and z_2 respectively:

$$Prob(C_{ji} < z_1 \text{ and } C_{j2} < z_2) = \phi \left(\frac{z_1 - \beta'_1 X_{j1}}{\sigma_1}, \frac{z_2 - \beta'_2 X_{j2}}{\sigma_2}, \rho \right) \quad (3)$$

4.1 Poverty Line: The \$8.30 per Day (2021 PPP) Threshold

As global living standards have risen, so too have the methods for measuring poverty. Successive revisions to the international poverty line implemented in 2001, 2008, 2015, 2022, and most recently this month reflect improvements in consumption measurement, updates to national poverty thresholds, and the use of Purchasing Power

Parities (PPPs) to harmonize costs and incomes across countries. The current update, which also affects middle-income country lines, draws on new PPPs from the 2021 International Comparison Program and incorporates recent changes in national poverty benchmarks a key factor behind the notable rise in the extreme poverty threshold. Based on these revised factors, the international poverty line for low-income countries is now up from \$2.15 to \$3 per person per day. For lower middle-income countries its changed from \$3.65 to \$4.15 per day and for upper middle-income countries it went up from \$6.85 to \$8.30.

The \$8.30 per day threshold, measured in 2021 Purchasing Power Parity (PPP), now is a standardized international poverty line established by the World Bank for assessing poverty in upper-middle-income countries (Foster et al., 2025; World Bank, 2025). It is derived from national poverty lines expressed in 2021 PPP dollars; the median among upper-middle-income countries is \$8.29 and is rounded to \$8.30 (Foster et al., 2025, Section 3.1 and Table 1). This line is part of a tiered measurement system that also includes an extreme poverty line (\$3.00) and a lower-middle-income line (\$4.15), allowing for a more nuanced analysis of economic well-being across development contexts (Foster et al., 2025; World Bank, 2025).

The use of the 2021 PPP update, including the \$8.30 line, is methodologically justified for several reasons. First, it uses a *current price benchmark*: the 2021 PPPs incorporate more recent price data, while the revised poverty lines draw on updated national poverty lines (Foster et al., 2025). Second, it ensures *cross-country consistency*, creating a standardized welfare metric that allows for coherent global poverty aggregation and comparison (Foster et al., 2025). Third, it addresses a measurement gap in *higher development contexts*, where the extreme poverty line is less informative; the \$8.30 threshold provides a more relevant benchmark for upper-middle-income countries (World Bank, 2025). Finally, it aligns the analysis with the World Bank’s current global poverty reporting framework, while the corresponding 2017 PPP series remains available through the Poverty and Inequality Platform (World Bank, 2025).

Nonetheless, PPP-based measures have recognized limitations. The construction of PPPs is methodologically complex, and revisions can alter measured poverty levels and create discontinuities in historical poverty series (Deaton, 2010; Foster et al., 2025). Uniform international lines may also fail to reflect locally specific needs and national definitions of poverty; national poverty lines are therefore more appropriate for country-specific policy dialogue and targeting (World Bank, 2025). Consequently, the \$8.30 (2021 PPP) line is used in this study as an international benchmark and should be considered alongside national poverty lines and, where relevant, complementary multidimensional deprivation measures (World Bank, 2025; Alkire and Foster, 2011).

5 Results

The present study examines poverty dynamics in two consequent years of 2019 and 2020, with a secondary objective of validating the results of the synthetic panel approach against those of true panel data. Table 5.1 presents the results for all urban and rural households. Three specifications are considered, with varying assumptions on the correlation between error terms (ρ). Specification 1 assumes that $\rho = 1$ for the lower bounds and $\rho = 0$ for the upper bounds, representing the parametric equivalent of the nonparametric bounds. Specification 2 approximates

ρ with 0.8 and 0.2 for the lower and upper bounds, respectively. Specification 3 further narrows the bounds by approximating ρ with 0.7 and 0.3 for the lower and upper bounds, respectively. It is expected that each specification provides narrower bounds compared to the previous specification, however, this also increases the possibility that the true poverty dynamic rates from the panel data fall outside of the bounds.

The synthetic panel approach performs well in replicating true poverty dynamics, as shown in Table 5.1. The genuine panel rates consistently fall within the estimated lower and upper bounds across all specifications. Under the parametric Specification 3, for example, the true persistence rate households poor in both 2019 and 2020 lies between 8.8% and 11.8%. This finding is statistically proved by the nonparametric bound that ranges from 6.7% to 14.6%, this fraction of households are suffering from chronic poverty. Across all transition categories, the true rates fall within the estimated bounds, confirming the validity of the synthetic approach. The true panel data indicate that 9.4% of households remained poor in both periods (chronic poverty), 7.5% escaped poverty, and 9.9% fell into poverty the highest transition rate, reflecting substantial downward mobility likely driven by macroeconomic volatility, high inflation, and sanctions. These findings underscore the importance of distinguishing between chronic and transitory poverty and highlight the vulnerability of non-poor households, particularly within the shrinking middle class.

Table 5.1: Poverty dynamics for all rural and urban households

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	14.6	16.1	12.9	11.8	9.4 0.4	8.8	8.2	7.1	6.7
Poor, nonpoor Std. Dev.	1.1	0.9	4.1	5.1	7.5 0.5	8.1	8.7	9.9	9.8
Nonpoor, poor Std. Dev.	3.4	3.3	6.5	7.5	9.9 0.5	10.6	11.2	12.3	11.2
Nonpoor, nonpoor Std. Dev.	80.9	79.7	76.6	75.5	73.1 0.6	72.5	71.9	70.8	72.3
N	5799	5799	5799	5799	11485	5799	5799	5799	5799

Note: This table presents poverty dynamics between two consecutive years, 2019 and 2020. The HIES data come from a rotating survey that tracks three-quarters of households in subsequent waves. The "Truth" column reports poverty status calculated from the genuine panel, while Specifications 13 apply the DLLM synthetic methodology to estimate lower and upper bounds. Across all specifications, the estimated bounds consistently contain the true poverty transition rates. 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Household heads' ages are restricted to between 25 and 55 in the first survey round.

To further explore poverty dynamics in Iran, we extend the analysis to specific population subgroups. Table 5.2 focuses on female-headed households. The estimates indicate that these households face a higher probability of persistent poverty relative to other household types. This result is consistent with the established finding that women encounter greater poverty risk due to restricted

labor market participation and occupational segregation. Female-headed households often bear disproportionate caregiving burdens and, lacking access to remunerative employment, frequently depend on informal, part-time, or low-wage work (Yoosefi Lebni et al., 2020). These findings underscore the necessity of incorporating gender-sensitive approaches into poverty reduction policies.

Table 5.2: Poverty dynamics for female-headed households

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	18.8	21.5	18.6	17.5	14.8 1.7	14.0	13.3	11.9	12.4
Poor, nonpoor Std. Dev.	3.8	2.7	5.6	6.7	9.4 1.9	10.2	10.9	12.3	11.5
Nonpoor, poor Std. Dev.	7.8	5.1	8.0	9.1	11.9 1.9	12.6	13.3	14.7	14.2
Nonpoor, nonpoor Std. Dev.	69.6	70.7	67.8	66.7	63.9 2.6	63.2	62.5	61.1	61.9
N	441	441	441	441	862	441	441	441	441

Note: 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Household heads' ages are restricted to between 25 and 55 in the first survey round.

Table Table 5.2 presents poverty transition rates for female-headed households between 2019 and 2020. The true panel data reveal that 14.8% of female-headed households remained poor in both periods-substantially higher than the 9.4% rate observed for all households. This substantially higher chronic poverty rate reflects the structural barriers women face in accessing stable, well-paid employment (Yoosefi Lebni et al., 2020). Additionally, 11.9% of female-headed households fell into poverty during this period (compared to 9.9% overall), while only 63.9% remained non-poor in both years (versus 73.1% overall). These findings highlight the heightened vulnerability of female-headed households to poverty and underscore the need for gender-responsive poverty reduction policies that address both chronic deprivation and the risk of future impoverishment.

Education influences chronic poverty primarily through its effect on labor market opportunities and earning capacity. Higher educational attainment is widely recognized as a pathway to improved employability and greater income-earning potential (Bilenkisi et al., 2015). Table ?? reports poverty transition rates for households headed by individuals without a university degree. The results show that these households face a higher probability of persistent poverty than those with university-educated heads. These findings reinforce the critical role of investment in education and skill development in reducing poverty and fostering upward economic mobility.

Table 5.3: Poverty dynamics among households with non-university-educated heads

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	17.7	19.1	15.1	13.9	11.0 0.5	10.4	9.7	8.3	8.0
Poor, nonpoor Std. Dev.	0.9	0.8	4.7	5.9	8.8 0.5	9.5	10.2	11.5	11.3
Nonpoor, poor Std. Dev.	2.4	3.1	7.1	8.3	11.1 0.5	11.8	12.5	13.8	12.2
Nonpoor, nonpoor Std. Dev.	78.9	77.0	73.1	71.9	69.1 0.8	68.3	67.6	66.3	68.6
N	5172	5172	5172	5172	10201	5172	5172	5172	5172

Note: 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Household heads' ages are restricted to between 25 and 55 in the first survey round.

Table 5.4: Poverty dynamics for households with a university-educated head

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	1.9	2.4	1.7	1.5	0.9 0.4	0.7	0.6	0.4	0.3
Poor, nonpoor Std. Dev.	0.0	0.1	0.8	1.0	1.6 0.6	1.8	1.9	2.1	1.8
Nonpoor, poor Std. Dev.	2.5	2.8	3.5	3.7	4.3 0.9	4.5	4.6	4.8	4.1
Nonpoor, nonpoor Std. Dev.	95.6	94.7	94.1	93.8	93.2 1.1	93.0	92.9	92.7	93.8
N	664	664	664	664	1309	664	664	664	664

Note: This table illustrates that between 2019 and 2020. 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Specification 1 assumes that $\alpha = 1$ and $\beta = 0$ for the lower bounds and upper bounds respectively and is the parametric equivalence of the nonparametric bounds. Specification 2 approximates α with 0.8 and 0.2, and Specification 3 approximates α with 0.7 and 0.3 for the lower bounds and upper bounds respectively. The regressors in all regressions are the same as in Model 1 in Table 1 plus an urban/rural dummy variable. 5. Household heads' ages are restricted to between 25 and 55 in the first survey round.

The protective effect of education against chronic poverty is strikingly evident when comparing households by the head's educational attainment. Among households with non-university-educated heads (Table [Table 5.3](#)), 11.0% remained poor in both periods, and 11.1% fell into poverty between 2019 and 2020 rates significantly higher than the full-sample averages of 9.4% and 9.9%, respectively. By con-

trast, households with university-educated heads (Table ??) exhibit substantially lower poverty persistence (0.9%) and entry rates (4.3%), with 93.2% maintaining nonpoor status across both periods. These findings confirm that higher education serves as a critical safeguard against poverty, enhancing employability and income-earning potential while reducing vulnerability to economic shocks (Bilenkisi et al., 2015). The education gap in poverty outcomes a 10.1 percentage point difference in chronic poverty and a 24.1 percentage point gap in sustained non-poverty underscores the importance of investment in education and skills training as a strategy for poverty reduction and economic mobility.

Tables Table 5.5 and Table 5.6 present the poverty dynamics of urban and rural households separately. The results show that the probability of remaining in poverty is higher for rural households as compared to urban households. Additionally, the rate of poverty transition (moving into or out of poverty in 2020) is lower for urban households as compared to rural households. These findings suggest that rural households are disproportionately affected by poverty and may require targeted interventions to reduce poverty and improve economic mobility. The higher poverty transition rate and lower probability of remaining in poverty for urban households may be due to the greater economic opportunities and access to services in urban areas. Overall, these findings highlight the importance of considering the specific needs and challenges faced by rural households in poverty reduction policies and programs.

Table 5.5: Poverty dynamics of urban households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	8.2	10.3	7.8	7.1	5.3 0.5	4.8	4.3	3.5	2.8
Poor, nonpoor Std. Dev.	0.8	0.7	3.1	3.9	5.6 0.5	6.2	6.7	7.5	7.8
Nonpoor, poor Std. Dev.	2.7	2.8	5.3	6.1	7.8 0.6	8.4	8.8	9.6	8.1
Nonpoor, nonpoor Std. Dev.	88.3	86.2	83.7	82.9	81.3 0.8	80.6	80.2	79.4	81.3
N	2987	2987	2987	2987	5873	2987	2987	2987	2987

Note: This table illustrates that between 2019 and 2020. 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Specification 1 assumes that $\alpha = 1$ and $\beta = 0$ for the lower bounds and upper bounds respectively and is the parametric equivalence of the nonparametric bounds. Specification 2 approximates α with 0.8 and 0.2, and Specification 3 approximates with 0.7 and 0.3 for the lower bounds and upper bounds respectively. The regressors in all regressions are the same as in Model 1 in Table 1 plus an urban/rural dummy variable. 5. Household heads' ages are restricted to between 25 and 55 in the first survey round.

Table 5.6: Poverty dynamics of rural households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	32.9	33.4	28.3	26.7	22.1 0.8	21.6	20.6	18.6	18.5
Poor, nonpoor Std. Dev.	0.6	1.0	6.1	7.7	12.1 0.7	12.7	13.8	15.8	15.8
Nonpoor, poor Std. Dev.	5.4	5.4	10.5	12.1	16.7 0.9	17.1	18.2	20.2	19.8
Nonpoor, nonpoor Std. Dev.	61.1	60.3	55.2	53.5	49.1 1.0	48.5	47.5	45.5	45.9
N	2865	2865	2865	2865	5627	2865	2865	2865	2865

Note: This table illustrates that between 2019 and 2020....

6 Robustness Checks

To evaluate the robustness of our findings, we extend the analysis to cover multiple transition windows: 2020-2021, 2021-2022, 2022-2023, and 2023-2024. We also examine in this section robustness to the choice of subpopulation groups. This multi-period approach is essential in the Iranian context, where the economy has been characterized by sustained inflationary pressures and recurrent sanctions-induced shocks ([World Bank, 2023](#); [Eslami and Hoseini, 2024](#)). In such a highly volatile setting, single-period poverty transition estimates may be influenced by short-term fluctuations rather than reflecting enduring structural patterns ([Baulch and Hoddinott, 2000b](#)). By replicating the analysis across several consecutive years, we can ascertain whether the observed poverty dynamics—particularly the heightened chronic poverty among female-headed and less-educated households—remain consistent over time or are specific to particular economic episodes ([Barrett et al., 2016](#)). This robustness strategy strengthens the reliability of our inferences and offers a more detailed picture of poverty persistence and vulnerability in Iran’s unstable economic environment ([Ravallion, 2001](#); [Dercon, 2005a](#)).

6.1 Robustness of Findings Across Time

To assess the robustness of our findings, we extended the analysis to cover four consecutive transition periods: 2020-2021, 2021-2022, 2022-2023, and 2023-2024, which are reported in [Table A.1](#) to [Table D.6](#). This multi-period approach is particularly important in the Iranian context, where the economy has been subjected to persistent inflationary pressures and recurring sanctions-induced shocks. In such a volatile environment, poverty dynamics observed in a single year may reflect transitory fluctuations rather than enduring structural patterns. By examining poverty transitions across several consecutive periods, we can determine whether the observed relationships—such as the elevated chronic poverty among female-headed and non-college-educated households—are stable over time or merely artifacts of specific economic episodes.

The results consistently confirm our main findings across all periods. For the full sample, chronic poverty rates (Poor to Poor) range from 6.0% to 11.0% across the four periods, with the highest rate observed in 2020-2021 (11.0%) and the lowest in 2023-2024 (6.0%). This declining trend in chronic poverty is accompanied by

a corresponding increase in sustained non-poverty (Nonpoor to Nonpoor), which rises from 69.2% in 2020-2021 to 81.0% in 2023-2024, suggesting a gradual improvement in household welfare over the period-potentially reflecting the recovery from the immediate shocks of the early 2020s. However, poverty entry rates (Nonpoor to Poor) remain substantial, ranging from 6.0% to 11.1%, indicating that vulnerability to poverty persists despite overall improvements.

The subgroup analyses reinforce the robustness of our key findings. Female-headed households consistently exhibit the highest chronic poverty rates across all periods, ranging from 8.6% in 2023-2024 to 15.5% in 2020-2021, substantially exceeding the full-sample rates. Similarly, households with non-university-educated heads show persistently higher chronic poverty (ranging from 6.9% to 12.7%) and lower sustained non-poverty (ranging from 64.8% to 78.1%) compared to their college-educated counterparts, whose chronic poverty rates remain near zero across all periods (0.5%-1.5%). The urban-rural divide is also strikingly consistent: rural households face chronic poverty rates three to five times higher than urban households across all periods (ranging from 15.1%-24.4% for rural vs. 2.8%-6.3% for urban). This persistent urban-rural gap underscores the structural disadvantages faced by rural populations, which are not eroded even in periods of overall improvement.

Importantly, the synthetic panel methodology performs well across all periods and subgroups. The true panel rates consistently fall within the lower and upper bounds estimated by the parametric and nonparametric specifications, confirming the reliability of the synthetic approach for studying poverty dynamics in Iran when genuine panel data are unavailable. The consistency of this validation across multiple years strengthens our confidence in both the methodology and the substantive findings.

Overall, the robustness checks across four transition periods confirm that our core conclusions are not driven by year-specific shocks or measurement artifacts. The elevated chronic poverty among female-headed and less-educated households, the large urban-rural disparity, and the protective effect of higher education against poverty persistence are consistent patterns that hold across Iran's volatile economic landscape. These findings provide a stable empirical foundation for policy recommendations aimed at reducing poverty and vulnerability in Iran.

Table 6.1: Key poverty transition rates across periods, by subgroup

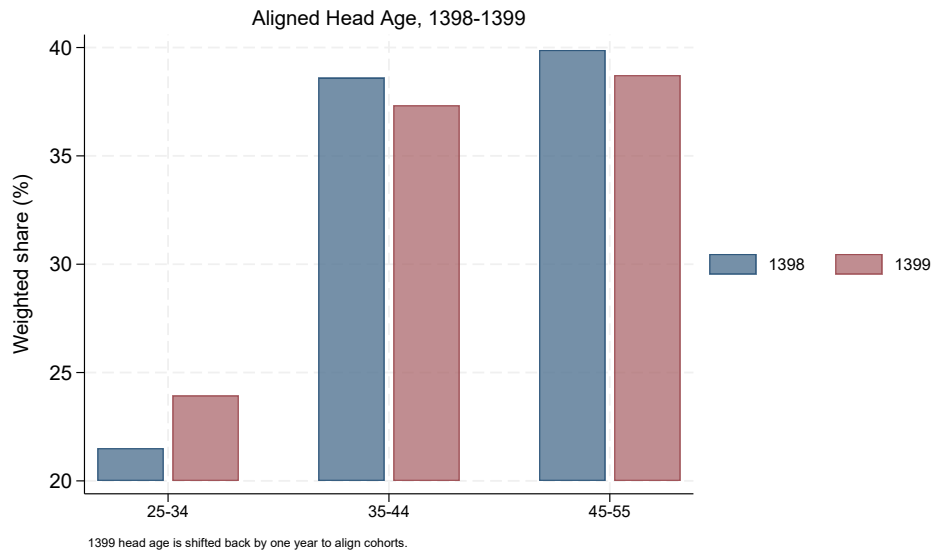
Subgroup	Indicator	2020-2021	2021-2022	2022-2023	2023-2024
Full Sample	Poor to Poor	11.0	9.7	7.4	6.0
	Nonpoor to Nonpoor	69.2	69.5	74.7	81.0
Female-Headed	Poor to Poor	15.5	15.2	10.4	8.6
	Nonpoor to Nonpoor	64.3	62.6	71.5	76.6
Non-College	Poor to Poor	12.7	10.8	8.6	6.9
	Nonpoor to Nonpoor	64.8	65.8	71.4	78.1
College	Poor to Poor	1.1	1.5	0.7	0.5
	Nonpoor to Nonpoor	92.4	90.7	93.9	96.4
Urban	Poor to Poor	6.3	5.3	3.8	2.8
	Nonpoor to Nonpoor	77.8	78.0	82.6	88.0
Rural	Poor to Poor	24.4	21.9	17.5	15.1
	Nonpoor to Nonpoor	45.1	45.8	53.1	61.3

Note: All rates are in percent. Source: HIES data, Statistical Center of Iran.

6.2 Composition Stability of the Synthetic Cohort

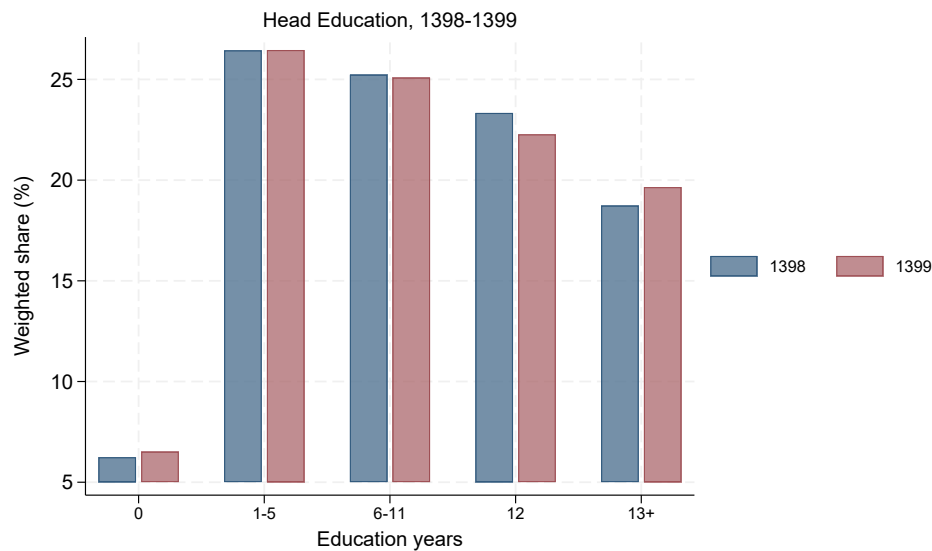
A key assumption in the synthetic-panel approach is that the two repeated cross-sections are drawn from comparable underlying populations. As a first diagnostic, we compare the weighted composition of the all-household sample used for the 1398-1399 estimates. The age window is aligned across rounds by comparing household heads aged 25–55 in 1398 with heads aged 26–56 in 1399. Figures 6.1 and 6.2 show that the weighted distributions of aligned head age groups and head education are broadly stable across the two survey rounds.

Figure 6.1: Weighted Distribution of Aligned Household-Head Age Groups, 1398-1399



Note: The sample follows the all-household 1398-1399 specification. The 1399 head age is shifted back by one year to align the synthetic cohort with 1398.

Figure 6.2: Weighted Distribution of Household-Head Education, 1398-1399

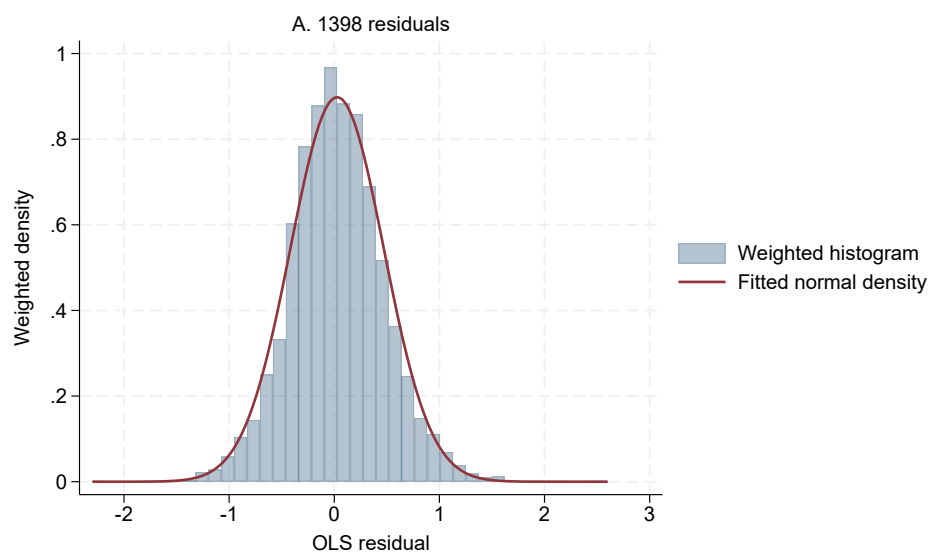


Note: Bars report weighted shares for the all-household age-aligned sample.

6.3 Residual Normality in the Consumption Models

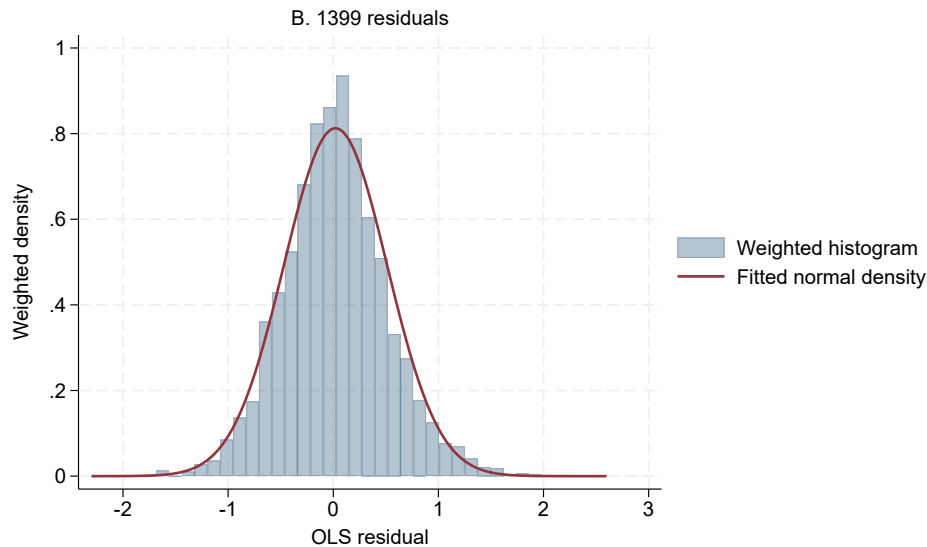
The parametric version of the synthetic-panel estimator relies on a distributional approximation for the regression errors. Following [Dang et al. \(2014\)](#), we visually assess this assumption by comparing residuals from the all-household 1398 and 1399 consumption regressions with fitted normal densities. Figures 6.3 and 6.4 show the two diagnostic plots separately so that the residual distributions can be inspected more clearly.

Figure 6.3: OLS Residuals and Fitted Normal Density, 1398



Note: Residuals are from the unweighted all-household OLS consumption model used for the 1398-1399 synthetic-panel estimates. The displayed density is weighted by HIES survey weights.

Figure 6.4: OLS Residuals and Fitted Normal Density, 1399



Note: Residuals are from the unweighted all-household OLS consumption model used for the 1398-1399 synthetic-panel estimates. The displayed density is weighted by HIES survey weights.

7 Potential Mechanisms

The observed patterns in poverty dynamics - particularly the higher rates of chronic poverty among female-headed, less-educated, and rural households - can be understood through several interrelated economic and sociological mechanisms, grounded in the literature on poverty traps and vulnerability. **we need a histogram for non-labor earnings of households based on the households with higher and less than diploma educated heads and the household head's gender for two distinct years.**

First, the asset-based poverty trap framework, articulated by [Barrett and Carter \(2008\)](#), provides a foundational explanation. Households with limited initial endowments of physical, financial, or human capital may fall below a critical threshold, preventing investment in income-generating activities and locking them into persistent poverty. This mechanism is evident in our findings: households headed by individuals without university education lack the human capital necessary to access stable, formal-sector employment, consistent with the human capital theory of poverty ([Schultz, 1961](#)). This lack of education constrains occupational mobility and resilience to economic shocks, as noted in studies from developing contexts ([Behrman, 1999](#)).

Second, structural and labor market inequalities drive the gendered and spatial dimensions of our results. Female-headed households face a compounded disadvantage due to occupational segregation, wage discrimination, and the disproportionate burden of unpaid care work ([World Bank, 2012](#)). Their overrepresentation in the informal sector, characterized by low pay and job insecurity ([ILO, 2020](#)), limits their capacity to accumulate savings or assets, increasing vulnerability to chronic poverty. This aligns with feminist political economy critiques that highlight how labor markets reproduce gendered poverty ([Elson, 1999](#)).

Third, the stark rural-urban divide can be attributed to spatial poverty traps. Rural households often depend on climate-sensitive agriculture, face higher trans-

portation costs, and have limited access to financial services, quality education, and non-farm employment (Jalan and Ravallion, 2002). This geographic disadvantage is exacerbated by infrastructure deficits and public service concentration in urban areas, a common feature in middle-income countries (Kanbur and Zhuang, 2014). The lower mobility out of poverty in rural areas reflects this structural constraint on economic diversification and opportunity.

Fourth, the broader macroeconomic and institutional context of Iran during the study period (2019-2020) serves as a critical precipitating mechanism. The confluence of stringent economic sanctions and the COVID-19 pandemic induced severe inflationary pressures, currency depreciation, and contraction in formal employment (Amid and Hadizadeh, 2021). Such macroeconomic shocks disproportionately affect the most vulnerable group—those with precarious employment, limited savings, and less diversified income sources—pushing transient poor households into chronic poverty and making escape more difficult for those already poor (Dercon, 2005b). The shock likely eroded the modest gains of those near the poverty line, a phenomenon documented in other crisis settings (Hallegatte et al., 2016).

Finally, the limited mobility observed, especially among the vulnerable subgroups, may reflect weak social protection systems. Effective safety nets and active labor market policies can mitigate shocks and facilitate poverty exits (Barrientos, 2011). The insufficiency or limited coverage of such programs in Iran during this period may have left households exposed to income volatility without adequate buffers, reinforcing chronic poverty (Moradi and Salehi-Isfahani, 2021).

In summary, the mechanisms driving our results are multifaceted, operating at the intersection of individual human capital deficits, structural market inequalities, geographic disparities, and adverse macroeconomic shocks. These factors interact to create and sustain the poverty traps evident in the data, particularly for the most marginalized segments of Iranian society.

8 Conclusion

In this research, we have presented a method for estimating poverty dynamics in the absence of suitable panel data. Specifically, we have employed the synthetic panel model proposed by Dang et al. (2014) to estimate the first-round consumption of households surveyed in the second round. We have applied this method to data from the Expenditures and Income Survey (HIES) for the years 2019 and 2020 in Iran, which is a period of particular significance due to the COVID-19 pandemic and US sanctions. Our results indicate that a significant proportion of households, particularly female-headed households, households with non-university educated heads, and rural households, are trapped in chronic poverty. Furthermore, we have also found that the estimates of poverty dynamics obtained from our synthetic panel model are in close agreement with those obtained from true panel data, demonstrating the validity and usefulness of this approach. Overall, this research highlights the importance of measuring poverty dynamics, which can provide valuable information for the design and implementation of poverty reduction policies.

Declaration

During the preparation of this work, the authors used DeepSeek in order to improve grammatical precision. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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A 2020-2021

Table A.1: Poverty dynamics for all rural and urban households

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	17.9	18.7	15.2	14.0	11.0 0.4	10.7	10.0	8.7	8.6
Poor, nonpoor Std. Dev.	0.9	1.0	4.5	5.7	8.7 0.5	9.0	9.7	11.0	10.8
Nonpoor, poor Std. Dev.	3.2	3.5	7.0	8.1	11.1 0.6	11.5	12.2	13.4	12.6
Nonpoor, nonpoor Std. Dev.	78.0	76.8	73.3	72.2	69.2 0.7	68.8	68.1	66.9	68.1
N	5593	5593	5593	5593	11142	5593	5593	5593	5593

Note: 2020-2021 (1399-1400), Source: HIES. Subpopulation: whole sample

Table A.2: Poverty dynamics for female-headed households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	16.9	21.2	18.7	17.8	15.5 1.4	14.7	14.1	12.9	11.9
Poor, nonpoor Std. Dev.	4.8	3.9	6.3	7.3	9.7 1.8	10.4	11.0	12.2	12.7
Nonpoor, poor Std. Dev.	5.2	4.8	7.3	8.2	10.5 1.6	11.3	11.9	13.1	10.2
Nonpoor, nonpoor Std. Dev.	73.1	70.1	67.7	66.7	64.3 2.4	63.6	63.0	61.8	65.2
N	459	459	459	459	874	459	459	459	459

Note: 2020-2021 (1399-1400), Source: HIES. Subpopulation: Female headed households

Table A.3: Poverty dynamics for households whose head lacks university education.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	19.9	21.5	17.5	16.2	12.7 0.5	12.4	11.6	10.1	9.9
Poor, nonpoor Std. Dev.	1.2	1.1	5.1	6.4	9.9 0.6	10.3	11.0	12.5	12.5
Nonpoor, poor Std. Dev.	3.7	3.8	7.8	9.1	12.6 0.6	12.9	13.7	15.1	13.7
Nonpoor, nonpoor Std. Dev.	75.3	73.6	69.6	68.3	64.8 0.9	64.5	63.7	62.2	63.9
N	4969	4969	4969	4969	9785	4969	4969	4969	4969

Note: 2020-2021 (1399-1400), Source: HIES. Subpopulation: households' head with non-college education

Table A.4: Poverty dynamics for households with a university-educated head

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	2.0	2.8	2.1	1.8	1.1 0.4	1.0	0.9	0.6	0.5
Poor, nonpoor Std. Dev.	0.3	0.3	1.0	1.3	1.9 0.6	2.0	2.2	2.4	2.3
Nonpoor, poor Std. Dev.	2.5	2.8	3.6	3.8	4.5 1.0	4.6	4.8	5.0	4.0
Nonpoor, nonpoor Std. Dev.	95.2	94.1	93.4	93.1	92.4 1.2	92.3	92.2	91.9	93.2
N	664	664	664	664	1346	664	664	664	664

Note: 2020-2021 (1399-1400), Source: HIES. Subpopulation: households' head with college education

Table A.5: Poverty dynamics of urban households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	10.4	12.1	9.5	8.6	6.3 0.5	6.0	5.5	4.5	4.0
Poor, nonpoor Std. Dev.	1.4	1.3	3.9	4.8	7.1 0.6	7.4	7.9	8.8	9.1
Nonpoor, poor Std. Dev.	2.5	3.0	5.6	6.5	8.8 0.6	9.1	9.6	10.5	8.9
Nonpoor, nonpoor Std. Dev.	85.7	83.7	81.0	80.1	77.8 0.8	77.5	77.0	76.1	78.0
N	2865	2865	2865	2865	5616	2865	2865	2865	2865

Note: This table illustrates that between 2020 and 2021. 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Specification 1 assumes that $\alpha = 1$ and $\beta = 0$ for the lower bounds and upper bounds respectively and is the parametric equivalence of the nonparametric bounds. Specification 2 approximates α with 0.8 and 0.2, and Specification 3 approximates α with 0.7 and 0.3 for the lower bounds and upper bounds respectively. The regressors in all regressions are the same as in Model 1 in Table 1 plus an urban/rural dummy variable. 5. Household heads' ages are restricted to between 25 and 55 in the first survey round.

Table A.6: Poverty dynamics of rural households

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	36.5	36.3	31.3	29.6	24.4 0.8	24.4	23.3	21.2	21.9
Poor, nonpoor Std. Dev.	2.6	2.5	7.5	9.2	14.2 0.8	14.4	15.5	17.6	17.2
Nonpoor, poor Std. Dev.	4.5	4.4	9.4	11.1	16.2 0.8	16.3	17.4	19.5	19.1
Nonpoor, nonpoor Std. Dev.	56.4	56.8	51.8	50.1	45.1 1.0	44.9	43.8	41.7	41.8
N	2832	2832	2832	2832	5526	2832	2832	2832	2832

Note: 2020-2021 (1399-1400), Source: HIES. Subpopulation: Rural area

B 2021-2022

Table B.1: Poverty dynamics for all rural and urban households

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	17.0	18.0	14.4	13.3	9.7 0.4	9.9	9.2	8.0	7.8
Poor, nonpoor Std. Dev.	3.3	3.3	7.0	8.1	11.6 0.6	11.4	12.1	13.3	13.1
Nonpoor, poor Std. Dev.	0.6	0.9	4.5	5.7	9.2 0.5	9.0	9.7	10.9	9.7
Nonpoor, nonpoor Std. Dev.	79.1	77.7	74.1	73.0	69.5 0.7	69.6	69.0	67.7	69.3
N	5582	5582	5582	5582	11008	5582	5582	5582	5582

Note: 2021-2022 (1400-1401), Source: HIES. Subpopulation: Whole sample

Table B.2: Poverty dynamics for female-headed households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	21.6	21.9	19.3	18.2	15.2 1.6	15.0	14.3	13.1	12.5
Poor, nonpoor Std. Dev.	7.3	7.1	9.7	10.8	13.7 2.0	14.0	14.7	15.9	16.0
Nonpoor, poor Std. Dev.	1.4	1.7	4.4	5.4	8.4 1.7	8.6	9.3	10.6	10.4
Nonpoor, nonpoor Std. Dev.	69.7	69.3	66.6	65.6	62.6 2.5	62.4	61.7	60.4	61.0
N	439	439	439	439	849	439	439	439	439

Note: 1400-1401, 2021-2022. This table illustrates that between 2021 and 2022. 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Specification 1 assumes that $\alpha = 1$ and $\beta = 0$ for the lower bounds and upper bounds respectively and is the parametric equivalence of the nonparametric bounds. Specification 2 approximates α with 0.8 and 0.2, and Specification 3 approximates α with 0.7 and 0.3 for the lower bounds and upper bounds respectively. The regressors in all regressions are the same as in Model 1 in Table 1 plus an urban/rural dummy variable. 5. Household heads' ages are restricted to between 25 and 55 in the first survey round.

Table B.3: Poverty dynamics for households whose head lacks university education

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	18.0	19.8	16.1	14.9	10.8 0.5	11.2	10.4	9.0	8.6
Poor, nonpoor Std. Dev.	5.0	4.9	8.7	9.9	13.9 0.7	13.6	14.4	15.7	15.8
Nonpoor, poor Std. Dev.	0.5	0.5	4.2	5.5	9.4 0.6	9.2	9.9	11.3	10.0
Nonpoor, nonpoor Std. Dev.	76.4	74.7	71.0	69.8	65.8 0.8	66.1	65.3	63.9	65.7
N	4871	4871	4871	4871	9650	4871	4871	4871	4871

Note: 1400-1401.

Table B.4: Poverty dynamics for households with a university-educated head

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	2.9	3.9	2.8	2.4	1.5 0.5	1.4	1.2	0.9	0.9
Poor, nonpoor Std. Dev.	0.6	1.0	2.1	2.4	3.4 0.8	3.4	3.6	4.0	3.5
Nonpoor, poor Std. Dev.	0.7	2.0	3.1	3.5	4.4 0.9	4.5	4.7	5.0	2.8
Nonpoor, nonpoor Std. Dev.	95.7	93.1	92.0	91.7	90.7 1.2	90.6	90.5	90.1	92.8
N	720	720	720	720	1362	720	720	720	720

Note: 1400-1401. This table illustrates that between 2021 and 2022. 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Specification 1 assumes that $\alpha = 1$ and $\beta = 0$ for the lower bounds and upper bounds respectively and is the parametric equivalence of the nonparametric bounds. Specification 2 approximates α with 0.8 and 0.2, and Specification 3 approximates with 0.7 and 0.3 for the lower bounds and upper bounds respectively. The regressors in all regressions are the same as in Model 1 in Table 1 plus an urban/rural dummy variable. 5. Household heads' ages are restricted to between 25 and 55 in the first survey round.

Table B.5: Poverty dynamics of urban households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	9.5	11.7	8.9	8.0	5.3 0.5	5.4	4.8	3.9	3.3
Poor, nonpoor Std. Dev.	3.1	2.7	5.6	6.5	9.2 0.7	9.1	9.6	10.6	10.7
Nonpoor, poor Std. Dev.	0.9	1.3	4.1	5.0	7.6 0.7	7.6	8.2	9.1	7.1
Nonpoor, nonpoor Std. Dev.	86.5	84.3	81.4	80.5	78.0 0.9	77.9	77.4	76.5	78.9
N	2845	2845	2845	2845	5601	2845	2845	2845	2845

Note: 1400-1401. 2021-2022, Source: HIES. Subpopulation: Urban area

Table B.6: Poverty dynamics of rural households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	35.9	34.6	29.6	27.9	21.9 0.8	22.7	21.6	19.6	20.7
Poor, nonpoor Std. Dev.	5.8	5.6	10.6	12.3	18.3 0.9	17.5	18.5	20.6	19.7
Nonpoor, poor Std. Dev.	1.7	1.5	6.5	8.2	14.0 0.8	13.3	14.4	16.5	17.0
Nonpoor, nonpoor Std. Dev.	56.6	58.3	53.3	51.6	45.8 1.0	46.5	45.4	43.3	42.6
N	2754	2754	2754	2754	5402	2754	2754	2754	2754

Note: 1400-1401. 2021-2022, Source: HIES. Subpopulation: Rural area

Table B.7: Poverty dynamics for all rural and urban households

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	11.1	12.9	10.7	9.9	7.4 0.4	7.3	6.8	5.8	5.4
Poor, nonpoor Std. Dev.	5.8	6.1	8.3	9.2	11.7 0.6	11.8	12.3	13.3	13.3
Nonpoor, poor Std. Dev.	0.6	0.7	2.8	3.7	6.2 0.4	6.3	6.8	7.8	6.3
Nonpoor, nonpoor Std. Dev.	82.5	80.3	78.1	77.3	74.7 0.7	74.6	74.1	73.1	75.0
N	5448	5448	5448	5448	10755	5448	5448	5448	5448

Note: 1400-1401. This table illustrates that between 2021 and 2022. 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Specification 1 assumes that $\alpha = 1$ and $\beta = 0$ for the lower bounds and upper bounds respectively and is the parametric equivalence of the nonparametric bounds. Specification 2 approximates α with 0.8 and 0.2, and Specification 3 approximates with 0.7 and 0.3 for the lower bounds and upper bounds respectively. The regressors in all regressions are the same as in Model 1 in Table 1 plus an urban/rural dummy variable. 5. Household heads' ages are restricted to between 25 and 55 in the first survey round.

C 2022-2023

Table C.1: Poverty dynamics for female-headed households

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	14.8	15.2	13.1	12.3	10.4 1.3	9.7	9.2	8.2	8.2
Poor, nonpoor Std. Dev.	9.3	6.5	8.6	9.4	11.3 1.7	12.0	12.5	13.5	12.9
Nonpoor, poor Std. Dev.	2.1	1.9	4.0	4.8	6.8 1.4	7.4	7.9	8.9	8.7
Nonpoor, nonpoor Std. Dev.	73.9	76.4	74.3	73.5	71.5 2.2	70.9	70.4	69.4	70.2
N	460	460	460	460	880	460	460	460	460

Note: 1401-1402. This table illustrates that between 2022 and 2023. 1. Poverty rates in percent are calculated using HIES data, statistical Center of Iran, and predictions obtained using data in the second survey rounds. Full regression results are provided in Tables 2.1a and 2.1b in Appendix ??.

2. All calculations are weighted using population weights for each survey round. Standard errors in parentheses.

3. Number of replications for the estimates is 500.

4. Specification 1 assumes that $\alpha = 1$ and $\beta = 0$ for the lower bounds and upper bounds respectively and is the parametric equivalence of the nonparametric bounds. Specification 2 approximates α with 0.8 and 0.2, and Specification 3 approximates with 0.7 and 0.3 for the lower bounds and upper bounds respectively. The regressors in all regressions are the same as in Model 1 in Table 1 plus an urban/rural dummy variable. 5. Household heads' ages are restricted to between 25 and 55 in the first survey round.

Table C.2: Poverty dynamics for households whose head lacks university education

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	13.0	14.9	12.5	11.5	8.6 0.4	8.5	7.9	6.8	6.5
Poor, nonpoor Std. Dev.	6.5	6.8	9.3	10.3	13.2 0.6	13.3	13.9	15.0	14.8
Nonpoor, poor Std. Dev.	0.4	0.5	3.0	3.9	6.8 0.5	6.9	7.6	8.7	7.0
Nonpoor, nonpoor Std. Dev.	80.0	77.7	75.3	74.3	71.4 0.8	71.3	70.7	69.6	71.7
N	4789	4789	4789	4789	9416	4789	4789	4789	4789

Note: 1401-1402.

Table C.3: Poverty dynamics for households with a university-educated head

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	1.8	2.2	1.6	1.3	0.7 0.4	0.7	0.6	0.4	0.2
Poor, nonpoor Std. Dev.	1.1	1.9	2.6	2.8	3.4 0.8	3.4	3.5	3.7	3.6
Nonpoor, poor Std. Dev.	0.3	0.6	1.3	1.5	2.0 0.7	2.1	2.2	2.4	1.9
Nonpoor, nonpoor Std. Dev.	96.8	95.3	94.6	94.4	93.9 1.1	93.7	93.6	93.4	94.3
N	689	689	689	689	1348	689	689	689	689

Note: 1401-1402.

Table C.4: Poverty dynamics of urban households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	5.2	7.8	6.2	5.6	3.8 0.4	3.7	3.3	2.7	1.8
Poor, nonpoor Std. Dev.	5.0	5.4	6.9	7.6	9.3 0.7	9.4	9.8	10.5	11.0
Nonpoor, poor Std. Dev.	0.3	0.4	1.9	2.6	4.3 0.5	4.4	4.8	5.5	3.7
Nonpoor, nonpoor Std. Dev.	89.5	86.5	84.9	84.3	82.6 0.8	82.5	82.1	81.4	83.5
N	2834	2834	2834	2834	5526	2834	2834	2834	2834

Note: 1401-1402.

Table C.5: Poverty dynamics of rural households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	27.0	27.0	23.5	22.1	17.5 0.8	17.6	16.7	14.8	15.2
Poor, nonpoor Std. Dev.	10.0	9.0	12.5	13.9	18.5 0.9	18.4	19.3	21.2	20.8
Nonpoor, poor Std. Dev.	1.1	1.4	4.9	6.3	10.9 0.7	10.8	11.8	13.6	12.9
Nonpoor, nonpoor Std. Dev.	61.8	62.6	59.1	57.7	53.1 1.0	53.2	52.2	50.4	51.0
N	2665	2665	2665	2665	5226	2665	2665	2665	2665

Note: 1401-1402.

D 2023-2024

Table D.1: Poverty dynamics for all rural and urban households

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	9.9	11.1	8.5	7.7	6.0 0.3	5.5	5.1	4.3	4.1
Poor, nonpoor Std. Dev.	1.5	1.9	4.5	5.2	7.0 0.4	7.5	7.9	8.7	8.4
Nonpoor, poor Std. Dev.	0.6	0.9	3.4	4.2	6.0 0.4	6.5	6.9	7.7	6.4
Nonpoor, nonpoor Std. Dev.	88.0	86.2	83.6	82.8	81.0 0.6	80.6	80.1	79.3	81.1
N	5337	5337	5337	5337	10546	5337	5337	5337	5337

Note: 1402-1403. 2023-2024, Source: HIES. Subpopulation: whole sample

Table D.2: Poverty dynamics for female-headed households

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	11.7	13.4	11.0	10.2	8.6 1.1	7.7	7.3	6.4	5.7
Poor, nonpoor Std. Dev.	3.0	3.8	6.2	7.0	8.8 1.7	9.5	10.0	10.8	10.7
Nonpoor, poor Std. Dev.	0.9	1.2	3.6	4.4	6.0 1.4	6.9	7.4	8.3	6.9
Nonpoor, nonpoor Std. Dev.	84.4	81.6	79.2	78.4	76.6 2.2	75.9	75.4	74.5	76.7
N	477	477	477	477	876	477	477	477	477

Note: 1402-1403.

Table D.3: Poverty dynamics for households whose head lacks university education

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	11.7	13.1	9.9	9.0	6.9 0.4	6.3	5.8	4.9	4.9
Poor, nonpoor Std. Dev.	1.7	2.1	5.3	6.2	8.3 0.5	8.9	9.4	10.3	9.8
Nonpoor, poor Std. Dev.	0.7	0.6	3.8	4.7	6.8 0.5	7.4	7.9	8.8	7.5
Nonpoor, nonpoor Std. Dev.	85.9	84.2	81.0	80.1	78.1 0.7	77.4	76.9	76.0	77.8
N	4684	4684	4684	4684	9181	4684	4684	4684	4684

Note: 1402-1403. 2023-2024, Source: HIES. Subpopulation: Female Headed house holds

Table D.4: Poverty dynamics for households with a university-educated head

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	1.4	1.4	0.9	0.8	0.5 0.3	0.4	0.3	0.2	0.2
Poor, nonpoor Std. Dev.	0.4	0.5	1.0	1.2	1.5 0.5	1.6	1.6	1.7	1.6
Nonpoor, poor Std. Dev.	0.4	0.7	1.2	1.4	1.7 0.6	1.8	1.9	2.0	1.5
Nonpoor, nonpoor Std. Dev.	97.9	97.3	96.8	96.7	96.4 0.8	96.3	96.2	96.1	96.6
N	698	698	698	698	1385	698	698	698	698

Note: 1402-1403.

Table D.5: Poverty dynamics of urban households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	4.6	6.2	4.4	3.9	2.8 0.4	2.4	2.2	1.7	1.4
Poor, nonpoor Std. Dev.	1.4	1.8	3.5	4.0	5.1 0.5	5.5	5.8	6.2	6.2
Nonpoor, poor Std. Dev.	0.6	0.6	2.4	2.9	4.0 0.5	4.4	4.6	5.1	3.8
Nonpoor, nonpoor Std. Dev.	93.5	91.4	89.7	89.2	88.0 0.7	87.7	87.4	87.0	88.7
N	2743	2743	2743	2743	5471	2743	2743	2743	2743

Note: 1402-1403.

Table D.6: Poverty dynamics of rural households.

Poverty Status	Nonparametric bound	Parametric lower bound			Truth	Parametric upper bound			Nonparametric bound
		Spec. 1	Spec. 2	Spec. 3		Spec. 3	Spec. 2	Spec. 1	
Poor, poor Std. Dev.	24.4	24.6	20.1	18.7	15.1 0.8	14.6	13.7	12.1	12.3
Poor, nonpoor Std. Dev.	2.5	2.5	7.0	8.4	12.0 0.8	12.6	13.5	15.1	14.6
Nonpoor, poor Std. Dev.	1.9	2.1	6.6	8.0	11.6 0.7	12.2	13.0	14.6	13.9
Nonpoor, nonpoor Std. Dev.	71.3	70.8	66.3	64.9	61.3 1.0	60.7	59.8	58.2	59.2
N	2578	2578	2578	2578	5082	2578	2578	2578	2578

Note: 1402-1403.